

MINI ASSESSMENT

Assessment Details

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Q1. Handling API Rate Limits

A 429 response means that the API has received too many requests in a given period and is rate limiting the client. The application should not retry immediately in a tight loop. It should use patience, exponential backoff, the Retry-After header when available, and a retry ceiling so that repeated failures do not continue indefinitely.

```
import time
import requests

max_retries = 5
for attempt in range(max_retries):
    response = requests.get(url)
    if response.status_code != 429:
        break
    retry_after = response.headers.get("Retry-After")
    wait = int(retry_after) if retry_after else 2 ** attempt
    time.sleep(wait)
```

Q2. Handling Malformed API Responses

Calling `response.json()` can fail when the server returns HTML, an empty body, plain text, or a truncated response instead of valid JSON. The response should first be checked for a successful status and an appropriate content type. The JSON conversion should be protected with a try/except block, while a short raw-body preview can be logged for diagnosis. A controlled fallback such as an empty dictionary can prevent the whole batch process from crashing.

```
if response.ok and "application/json" in response.headers.get("Content-Type", ""):
    try:
        data = response.json()
    except ValueError:
        print(response.text[:200])
        data = {}
else:
    data = {}
```

Q3. Safe Dictionary Access

Using square brackets such as `data['field']` raises a `KeyError` when the requested key is missing. Using `data.get('field')` returns `None` when the key is absent, or it can return a supplied default value. This approach is useful when an API schema changes or when batch processing must continue even when some responses do not contain every expected field.

```
value = data.get("field")
status = data.get("status", "unknown")

# Compared with:
value = data["field"] # may raise KeyError
```